Wikipedia Math Essentials as a Directed Knowledge Network

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GitHub Repo: `https://github.com/Manan-KK/Csci4022FinalProject`

YouTube Link: `https://youtu.be/x43xjcelQeg`

Abstract

This project studies the Wikipedia Math Essentials dataset as both a directed hyperlink graph and a daily page-visit time series. The main question is whether graph structure explains public attention: which math pages are central, whether central pages are also heavily visited, whether soft clusters reveal overlapping communities, and whether graph-derived features can predict average page visits. The final graph contains 1,068 pages and 27,079 directed weighted links over 731 daily visit snapshots from 2019-03-16 through 2021-03-15. PageRank identified structurally central pages that were often specialized rather than broadly popular. Gaussian mixture models found five soft clusters with high average assignment confidence, but also a set of ambiguous pages that behave like bridge topics. Regression results show that centrality and SVD features improve prediction over PageRank alone, but the best model explains only about 24.2% of held-out variation in log mean visits. Overall, Wikipedia link structure is informative, but it is not a direct measure of mathematical importance or reader demand.

1 Introduction and Research Questions

The proposal for this project framed Wikipedia mathematics as a network-analysis problem rather than a simple popularity ranking. Each page is a node, each hyperlink is a directed edge, and each daily snapshot records how often readers visited each page. This setup makes it possible to compare two different notions of importance: structural importance in the knowledge graph and public attention through page views.

The analysis addresses five research questions. First, which Wikipedia mathematics pages are most central in the directed graph? Second, are structurally central pages also the pages with more visits? Third, do soft clusters reveal overlapping mathematical communities? Fourth, do low-confidence cluster assignments identify bridge pages between areas? Fifth, can centrality and latent graph embeddings predict average page visits?

2 Data and Methods

The raw source is the Wikipedia Math Essentials dataset from the UCI Machine Learning Repository. The local JSON file contains an edge list, edge weights, a mapping from page titles to node indices, and daily visit vectors. Table 1 summarizes the final objects used in the analysis.

Table 1: Dataset and diagnostic summary.

Quantity	Value
Pages / nodes	1,068
Directed weighted edges	27,079
Daily visit snapshots	731
Date range	2019-03-16 to 2021-03-15
Isolated nodes	0
Weakly connected components	1
Strongly connected components	1,068
Chosen GMM clusters	5
Mean GMM confidence	0.917
Pages with confidence below 0.60	50

The graph was constructed as a directed weighted NetworkX graph. PageRank was computed with damping factor

$d = 0.85$:

$$PR(i) = \frac{1-d}{N} + d \sum_{j \to i} PR(j) \frac{w_{ji}}{\sum_k w_{jk}}.$$

For clustering, graph and visit features were transformed with $\log(1+x)$ when nonnegative and skewed, standardized as $z = (x-\mu)/\sigma$, and modeled with Gaussian mixture models. The GMM confidence for page i was defined as $\max_k p(c = k \mid z_i)$, and normalized entropy was used to identify ambiguous pages:

$$H_i = -\sum_k p_{ik} \log(p_{ik}), \qquad H_i^{norm} = H_i / \log(K).$$

For latent graph structure, the normalized adjacency matrix was embedded with truncated SVD, $A_{norm} \approx U_k \Sigma_k V_k^T$. Regression used the target $y_i = \log(1 + \text{mean visits}_i)$ and RidgeCV:

$$\min_\beta \|y - X\beta\|^2 + \alpha\|\beta\|^2.$$

The code was implemented in Python with pandas, NumPy, NetworkX, SciPy, scikit-learn, and matplotlib in the project notebook.

3 Results

3.1 Centrality and Attention

PageRank produced a different ranking than page visits. The top PageRank pages included specialized graph destinations such as *Rotations in 4-dimensional Euclidean space*, *24-cell*, *Symmetric relation*, and *Branch point*. In contrast, the highest-traffic pages were broad or classroom-facing topics such as *Normal distribution*, *Standard deviation*, *Fibonacci number*, *Pi*, *Roman numerals*, and *Prime number*.

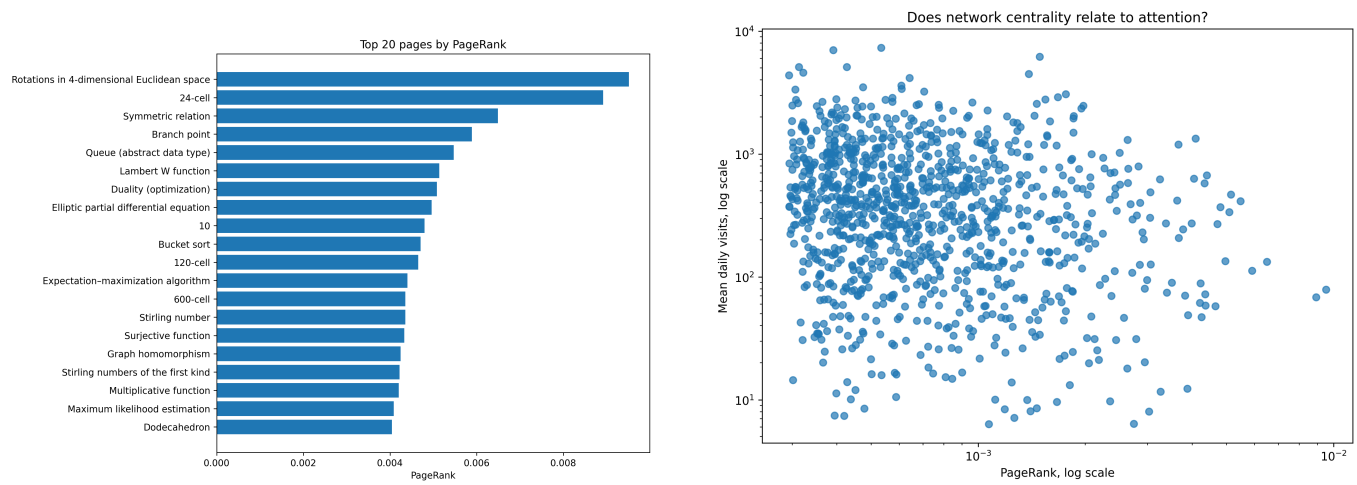

Figure 1: Left: top pages by PageRank. Right: PageRank versus mean visits on log-log scales. Structurally central pages are not necessarily the same pages that attract broad public attention.

The correlation between PageRank and mean visits was negative and weak ($r = -0.129$). In-degree and weighted in-degree were more positively related to visits ($r = 0.158$ and $r = 0.188$), but still not strong. This supports the main conclusion that link authority and reader demand are different signals.

3.2 Soft Clustering and Bridge Pages

The GMM model selection curve suggested a bend around $k = 4$ or $k = 5$, and $k = 5$ was chosen because it preserved interpretability while giving more structure for discussion. The clusters differ in size, PageRank, degree profiles, visits, and confidence. Cluster 0 is small but has the highest average PageRank, while clusters 2 and 3 contain broader high-out-degree and higher-visit pages.

Table 2: GMM cluster diagnostics.

Cluster	Pages	Avg. PR	Avg. in	Avg. out	Avg. visits
0	84	0.00265	30.8	8.2	485
1	278	0.00096	30.1	13.6	599
2	300	0.00052	33.0	46.7	787
3	158	0.00037	12.0	51.5	850
4	248	0.00119	17.4	1.8	291

Soft clustering was useful because the low-confidence pages were often conceptually mixed. Examples include *Even and odd functions*, *Right angle*, *Field extension*, *Harmonic analysis*, and *Fixed point (mathematics)*. These pages are plausible bridges because they connect definitions, algebraic structure, geometry, analysis, and computation rather than belonging cleanly to only one area.

3.3 SVD Interpretation and Problems

SVD was useful for creating latent graph features, but it was not useful as a standalone visualization. The first two-dimensional SVD projection was dominated by outliers and many points near zero, making the scatterplot visually uninformative. The component signs are also arbitrary, and components should not be treated as automatic subject labels. A more reliable interpretation came from inspecting the pages with the largest positive and negative component values.

Table 3: Examples of pages anchoring the first two SVD directions.

Component	Positive extreme examples	Negative / opposite examples
SVD 1	KKT conditions; Jensen's inequality; Transportation theory	Blum's speedup theorem; Five lemma; Snake lemma
SVD 2	Möbius function; Additive function; Dirichlet character	KKT conditions; Jensen's inequality; Gauss–Seidel method
SVD 3	Dirichlet problem; Poisson's equation; Elliptic operator	Möbius function; Additive function; Dirichlet character

This table suggests that SVD 1 is strongly tied to optimization and programming-related pages, SVD 2 to number theory, and SVD 3 to differential-equation or PDE-related pages. However, these interpretations are descriptive and require manual inspection. The SVD-only regression also performed poorly, which suggests that latent link structure by itself was not enough to explain attention.

3.4 Regression

The regression comparison confirms that PageRank alone is a weak predictor of visits. Centrality features improved performance, SVD alone did not, and the best model combined centrality with SVD features.

Table 4: Held-out regression results for predicting $\log(1 + \text{mean visits})$.

Model	R^2	MAE	RMSE
PageRank only	0.021	1.006	1.248
Centrality only	0.189	0.908	1.136
SVD only	0.045	0.992	1.232
Centrality + SVD	0.242	0.864	1.098

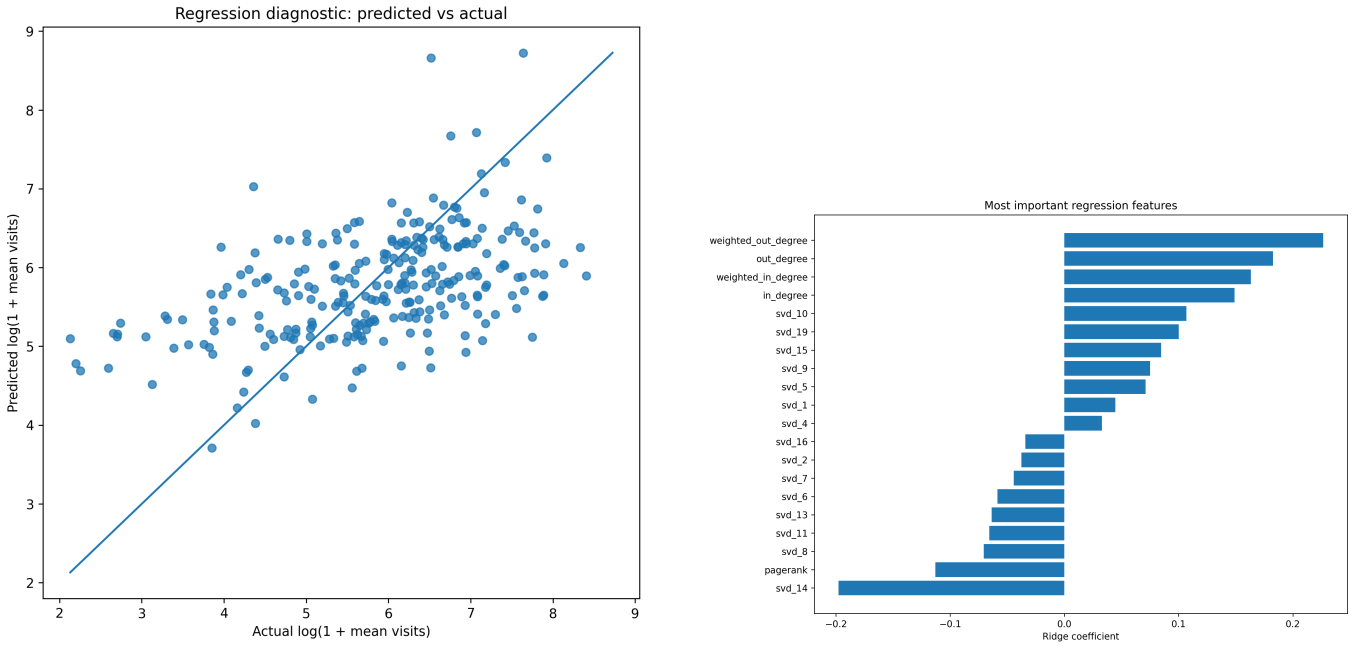

Figure 2: Regression diagnostics. The best model follows the correct direction but leaves wide residual variation. The largest positive coefficients include weighted out-degree, out-degree, weighted in-degree, and in-degree.

The best model achieved $R^2 = 0.242$, so graph structure explains some attention but leaves most variation unexplained. The largest positive coefficient was weighted out-degree, followed by out-degree, weighted in-degree, and in-degree. PageRank had a negative coefficient after other graph features entered the model, reinforcing that PageRank is not the same as public readership.

4 Limitations and Future Work

The main limitation is that Wikipedia is not a neutral map of mathematics. Links are created by editors, not by a formal ontology, and page-view counts reflect public curiosity, course calendars, search behavior, and internet events rather than mathematical importance. Broad pages such as *Normal distribution* or *Pi* are easier for general readers to find than specialized pages, so attention is biased toward public-facing topics. The time period from 2019 to March 2021 may also include unusual patterns from school schedules and pandemic-era online learning.

There are also modeling limitations. The directed graph is weakly connected but not strongly connected, so many links act more like outward references than reciprocal relationships. SVD components are difficult to interpret without inspecting page extremes, and the two-dimensional SVD scatterplot can be misleading because it compresses a high-dimensional graph into a small view. GMM clusters are useful as soft groupings, but the cluster labels are not true mathematical fields unless supported by page inspection. Finally, the regression performance shows that the graph features are incomplete predictors of visits.

Future work should add richer text features from page content, compare multiple graph embeddings, include temporal models for visits, and test whether seasonal or school-calendar effects explain traffic spikes. A more formal validation step could compare cluster results against human-labeled mathematical areas.

5 Conclusion

The project shows that Wikipedia mathematics has at least two different maps: a structural map from hyperlinks and an attention map from visits. PageRank identifies specialized authority sinks, while visits favor broad educational topics. Soft clustering reveals communities and ambiguous bridge pages, and regression shows that centrality plus latent graph structure predicts more attention than PageRank alone. At the same time, the modest R^2 and the SVD interpretation problems show that graph structure is only part of the story. Wikipedia is a useful observational dataset, but not a perfect proxy for mathematical importance.

References

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